The Minimum (a, b)-Additive Chromatic Violation Problem

Introduction

A *lucky labeling* (or *additive coloring*) of a graph G is a labeling $f : V \to \{1, \ldots, k\}$ such that $f(N(u)) \neq f(N(v))$ for every edge uv, where $f(S)$ is the sum of the labels of the vertices of S; i.e. the sums of the labels of the neighbors form a proper coloring of G. The least such k is the *lucky number* (or *additive chromatic number*) $\eta(G)$. Czerwiński, Grytczuk and Żelazny [3] introduced this parameter and proposed the following conjecture.

Additive Coloring Conjecture [3]. For every graph G, $\eta(G) \leq \chi(G)$.

The conjecture has been proved for some families, such as trees, complete graphs, some planar graphs of large girth, and split graphs (see [9] and the references therein). It is open even for bipartite graphs. There, since $\chi(G) = 2$ when G has at least one edge, the conjecture states that $\eta(G) \leq 2$. This has been proved for some subclasses, for example trees [3] and regular bipartite graphs [9]. For planar bipartite graphs, Czerwiński et al. proved $\eta(G) \leq 3$ [3], and, as far as we know, it is not even known whether η is bounded on bipartite graphs.

In the terminology of the next section, $\eta(G) \leq 2$ means that G has a $(1, 2)$-additive coloring, since a labeling with labels in $\{1\}$ is in particular a labeling with labels in $\{1, 2\}$. That is, $\eta(G) \leq 2$ if and only if $\eta_v^{1,2}(G) = 0$. Therefore, the Additive Coloring Conjecture restricted to bipartite graphs is equivalent to the statement

$$\eta_v^{1,2}(G) = 0 \qquad \text{for every bipartite graph } G,$$

and a bipartite graph with $\eta_v^{1,2}(G) > 0$ would be a counterexample. This is one motivation for studying the minimization problem below and the polytope of its integer programming formulation. Note that only the ratio $a : b$ matters (Remark 1), so $(1, 2)$ can be replaced by any pair $(t, 2t)$ with $t > 0$.

1 Definitions

Let $G = (V, E)$ be a simple graph, $N(v)$ the set of neighbors of v and $d(v) = |N(v)|$ its degree. For a function $g : V \to \mathbb{Z}$ and $S \subseteq V$ we write $g(S) = \sum_{w \in S} g(w)$. Throughout, a and b are integers with $a < b$, and $\delta = b - a > 0$. (The case $a > b$ is the same problem with the roles of a and b exchanged, and the case $a = b$ is trivial, since then the colors do not depend on the labeling.)

Definition 1. An (a, b)-*labeling* of G is a function $\ell : V \to \{a, b\}$. Its *color function* $c_\ell : V \to \mathbb{Z}$ is $c_\ell(v) = \ell(N(v))$. An edge $uv \in E$ is *violated* by ℓ if $c_\ell(u) = c_\ell(v)$, and the *violation* of ℓ is

$$\nu_{a,b}(G, \ell) = |\{uv \in E : c_\ell(u) = c_\ell(v)\}|.$$

The labeling ℓ is an (a, b)-*additive coloring* of G if $\nu_{a,b}(G, \ell) = 0$, i.e. if c_ℓ is a proper coloring of G.

Definition 2. The *Minimum* (a, b)-*Additive Chromatic Violation Problem* consists of finding an (a, b)-labeling of G of minimum violation. Its optimal value is denoted by

$$\eta_v^{a,b}(G) = \min\{\nu_{a,b}(G, \ell) : \ell : V \to \{a, b\}\}.$$

Clearly, $\eta_v^{a,b}(G) = 0$ if and only if G admits an (a,b)-additive coloring. The case $(a,b) = (0,1)$ corresponds to the $(0,1)$-additive labelings of Ahadi and Dehghan [2] (who denote by $\eta_1(G)$ the minimum number of 1's in a $(0,1)$-additive labeling, a different parameter). The notation is borrowed from the chromatic violation number $\chi_v(G,F,C)$ of the minimum chromatic violation problem, introduced in [7]; the symbols $\chi_v(G,F,C)$ and $\nu(G,F,\varphi)$, the latter for the violation of a semi-coloring φ, are those of [8].

Remark 1. No bound is imposed on the number of colors used by c_ℓ. Moreover, $\eta_v^{a,b}(G) = \eta_v^{-b,-a}(G)$ (multiply ℓ by -1), and $\eta_v^{a,b}(G) = \eta_v^{ta,tb}(G)$ for every integer $t > 0$ (multiply ℓ by t: all colors are multiplied by t, so the violated edges do not change).

Proposition 1. *If G is the disjoint union of $G_1, \ldots, G_n$, then $\eta_v^{a,b}(G) = \sum_{i=1}^n \eta_v^{a,b}(G_i)$.*

Proof. An (a,b)-labeling ℓ of G is the same as a choice of an (a,b)-labeling ℓ_i of each G_i, where ℓ_i is the restriction of ℓ to $V(G_i)$. If $v \in V(G_i)$, then $N(v) \subseteq V(G_i)$, so $c_\ell(v) = c_{\ell_i}(v)$. Since every edge of G belongs to some G_i, we get $\nu_{a,b}(G,\ell) = \sum_{i=1}^n \nu_{a,b}(G_i,\ell_i)$. The terms of this sum depend on independent choices, so its minimum over all ℓ is the sum of the minima, i.e. $\sum_{i=1}^n \eta_v^{a,b}(G_i)$. □

In particular, G admits an (a,b)-additive coloring if and only if each of its components does, and isolated vertices do not contribute to $\eta_v^{a,b}(G)$. Note the difference with the additive number, for which $\eta(G)$ is the maximum of the $\eta(G_i)$. From now on we assume that G is connected.

Example 1. Let G be the *net*: a triangle $t_1t_2t_3$ together with a pendant vertex p_i adjacent to t_i, for $i = 1,2,3$. G is additive 2-colorable and admits $(0,1)$- and $(1,2)$-additive colorings, but $\eta_v^{-1,1}(G) = 1$.

Indeed, let ℓ be a $(-1,1)$-labeling, $S = \ell(\{t_1,t_2,t_3\})$ and $d_i = \ell(p_i) - \ell(t_i) \in \{-2,0,2\}$. Then $c(p_i) = \ell(t_i)$ and $c(t_i) = S + d_i$. Hence the edge t_it_j is violated if and only if $d_i = d_j$, and the edge t_ip_i is violated if and only if $d_i = \ell(t_i) - S$. Suppose that no edge is violated. Then $\{d_1,d_2,d_3\} = \{-2,0,2\}$. If $\ell(t_1) = \ell(t_2) = \ell(t_3) = \epsilon$, then every $d_i \in \{0,-2\epsilon\}$, a contradiction. Otherwise $S = \pm 1$; by symmetry ($\ell \mapsto -\ell$) assume $S = 1$, so the labels of the triangle are $1,1,-1$. The vertex with label -1 has $d_i \in \{0,2\}$ and the two vertices with label 1 have $d_i \in \{-2,0\}$, so one of the latter has $d_i = 0 = \ell(t_i) - S$ and its pendant edge is violated, a contradiction. Therefore $\eta_v^{-1,1}(G) \geq 1$. The labeling $\ell(t_1,t_2,t_3) = (-1,-1,1)$, $\ell(p_1,p_2,p_3) = (-1,1,-1)$ gives $c(t_1,t_2,t_3) = (-1,1,-3)$ and $c(p_1,p_2,p_3) = (-1,-1,1)$, whose only violated edge is t_1p_1.

2 An integer programming formulation

Let $\overline{E} = \{(u,v),(v,u) : uv \in E\}$. For $(u,v) \in \overline{E}$ let

$$P_{uv} = N(u) \backslash N(v), \qquad Q_{uv} = N(v) \backslash N(u).$$

Note that $u \in Q_{uv}$, so $|Q_{uv}| \geq 1$.

We use binary variables $x(v)$ for $v \in V$, where the label of v is a if $x(v) = 0$ and b if $x(v) = 1$, i.e. $\ell(v) = a + \delta\, x(v)$, and binary variables $z(u,v)$ for $(u,v) \in \overline{E}$, where $z(u,v) = 1$ forces $c(u) < c(v)$. Since $c(u) = a\, d(u) + \delta\, x(N(u))$ and the common neighbors of u and v cancel out,

$$c(u) - c(v) = a\,(d(u) - d(v)) + \delta\,\big(x(P_{uv}) - x(Q_{uv})\big). \tag{1}$$

For $(u,v) \in \overline{E}$ define

$$\beta_{uv} = \left\lceil \frac{a\,(d(v) - d(u))}{\delta} \right\rceil - 1, \qquad M_{uv} = |P_{uv}| - \beta_{uv}.$$

The formulation is:

$$\begin{aligned}
&\min |E| - \sum_{(u,v)\in\overline{E}} z(u,v) \\
&\text{s.t. } x(P_{uv}) - x(Q_{uv}) + M_{uv}\, z(u,v) \le M_{uv} + \beta_{uv}, && \forall\ (u,v) \in \overline{E} && (2)\\
&\quad z(u,v) + z(v,u) \le 1, && \forall\ uv \in E && (3)\\
&\quad z(u,v) \in \{0,1\}, && \forall\ (u,v) \in \overline{E} && (4)\\
&\quad x(v) \in \{0,1\}, && \forall\ v \in V && (5)
\end{aligned}$$

Lemma 1. *Let $x \in \{0,1\}^V$ and $(u,v) \in \overline{E}$. Then $c(u) < c(v)$ if and only if $x(P_{uv}) - x(Q_{uv}) \le \beta_{uv}$.*

Proof. Let $D = x(P_{uv}) - x(Q_{uv}) \in \mathbb{Z}$. By (1), $c(u) < c(v)$ if and only if $D < a\,(d(v) - d(u))/\delta$. For an integer D and a real number r, $D < r$ if and only if $D \le \lceil r \rceil - 1$. □

Since $x(P_{uv}) - x(Q_{uv}) \le |P_{uv}| = M_{uv} + \beta_{uv}$, constraint (2) is redundant when $z(u,v) = 0$, and by Lemma 1 it is equivalent to $c(u) < c(v)$ when $z(u,v) = 1$. Therefore an integer vector (x,z) satisfies (2), (4) and (5) if and only if

$$z(u,v) = 1 \Longrightarrow c(u) < c(v), \qquad \forall\,(u,v) \in \overline{E}, \tag{6}$$

where c is the color function of the labeling encoded by x. In particular, constraints (3) are implied by the others on integer points, and (2) can be dropped whenever $M_{uv} \le 0$ (in that case $c(u) < c(v)$ holds for every x).

Proposition 2. *The optimal value of the formulation is $\eta_v^{a,b}(G)$.*

Proof. Let (x,z) be feasible and ℓ the labeling encoded by x. If $z(u,v) + z(v,u) = 1$ for an edge uv, then by (6) $c(u) \ne c(v)$ and uv is not violated. By (3), the objective value equals the number of edges uv with $z(u,v) = z(v,u) = 0$, which is at least $\nu_{a,b}(G,\ell) \ge \eta_v^{a,b}(G)$. Conversely, given an (a,b)-labeling ℓ, let $x(v) = 1$ if and only if $\ell(v) = b$, and $z(u,v) = 1$ if and only if $c(u) < c(v)$. This vector is feasible and its objective value is $\nu_{a,b}(G,\ell)$. □

3 Dimension of the polytope

Let $P^{a,b}(G) \subseteq \mathbb{R}^V \times \mathbb{R}^{\overline{E}}$ be the convex hull of the integer solutions of (2)–(5).

Proposition 3. *Let*

$$ST = \{(s,t) \in \overline{E} : b\,|N(t)\backslash N(s)| > a\,|N(s)\backslash N(t)|\}.$$

Then $\dim(P^{a,b}(G)) = |V| + |ST|$ and a minimal equation system for $P^{a,b}(G)$ is:

$$z(s,t) = 0, \qquad \forall\,(s,t) \in \overline{E}\backslash ST.$$

Proof. Let $(s,t) \in \overline{E}$ and $x \in \{0,1\}^V$. By (1), $c(t) - c(s) = \ell(Q_{st}) - \ell(P_{st}) \le b\,|Q_{st}| - a\,|P_{st}|$, with equality when x is the characteristic vector $\chi^{Q_{st}}$ of Q_{st} (recall that $P_{st} \cap Q_{st} = \emptyset$). Therefore

$$(s,t) \notin ST \iff c(s) \ge c(t) \text{ for every } x \in \{0,1\}^V \iff \chi^{Q_{st}} \text{ does not satisfy } c(s) < c(t). \tag{7}$$

By (6) and (7), every integer solution satisfies $z(s,t) = 0$ for all $(s,t) \notin ST$. Since these equalities are linearly independent, $\dim(P^{a,b}(G)) \le |V| + |ST|$.

To prove that $\dim(P^{a,b}(G)) \ge |V| + |ST|$, we propose $|V| + |ST| + 1$ affinely independent points of $P^{a,b}(G)$. They are feasible by (6) and (7).

- Sol. 0: $x^0 = \mathbf{0}$, $z^0 = \mathbf{0}$.
- Sol. r, for $r \in V$: $x^r = e_r$ (i.e. $\ell(r) = b$ and $\ell(w) = a$ for all $w \neq r$), $z^r = \mathbf{0}$.
- Sol. (s,t), for $(s,t) \in ST$: $x^{(s,t)} = \chi^{Q_{st}}$, $z^{(s,t)} = e_{(s,t)}$.

Consider coefficients α such that

$$\alpha^0(x^0, z^0) + \sum_{r \in V} \alpha^r(x^r, z^r) + \sum_{(s,t) \in ST} \alpha^{(s,t)}(x^{(s,t)}, z^{(s,t)}) = 0, \qquad (A)$$

$$\alpha^0 + \sum_{r \in V} \alpha^r + \sum_{(s,t) \in ST} \alpha^{(s,t)} = 0. \qquad (B)$$

The component $(s',t') \in ST$ of (A) gives $\alpha^{(s',t')} = 0$. Then the component $r' \in V$ of (A) gives $\alpha^{r'} = 0$, and finally $\alpha^0 = 0$ by (B). □

Remark 2. The proof does not require a and b to be consecutive integers. In the formulation with the original variables $f(v) = a + \delta\, x(v) \in \{a, b\}$, Sol. r has $f^r(r) = a + \delta = b$, and the component r' of (A) minus a times (B) gives $\delta\, \alpha^{r'} = 0$, hence $\alpha^{r'} = 0$ since $\delta \neq 0$. The only other place where the labels matter is the strict inequality $c(s) < c(t)$, which is equivalent to $c(s) \leq c(t) - 1$ because colors are integers.

Corollary 1. *If $a \leq 0 \leq b$ (in particular for $(a,b) = (0,1)$ and $(a,b) = (-1,1)$), then $P^{a,b}(G)$ is full-dimensional, i.e. $\dim(P^{a,b}(G)) = |V| + 2|E|$. For example, for the net G of Example 1, $\dim(P^{-1,1}(G)) = 6 + 12 = 18$.*

Proof. Let $(s,t) \in \overline{E}$. Then $|N(t) \backslash N(s)| \geq 1$ and $|N(s) \backslash N(t)| \geq 1$, since $s \in N(t) \backslash N(s)$ and $t \in N(s) \backslash N(t)$. Hence $b\,|N(t) \backslash N(s)| \geq 0 \geq a\,|N(s) \backslash N(t)|$, and at least one of these inequalities is strict because $a < b$ implies $a < 0$ or $b > 0$. Therefore $(s,t) \in ST$, and $ST = \overline{E}$. □

Remark 3. Outside the range of the corollary, ST may be a proper subset of $\overline{E}$. If $0 < a < b$, then $(s,t) \notin ST$ if and only if $|N(s) \backslash N(t)| \geq (b/a)\,|N(t) \backslash N(s)|$, i.e. s has many more private neighbors than t. If $a < b < 0$, the condition is mirrored: dividing $b\,|N(t) \backslash N(s)| \leq a\,|N(s) \backslash N(t)|$ by $b < 0$, $(s,t) \notin ST$ if and only if $|N(t) \backslash N(s)| \geq (a/b)\,|N(s) \backslash N(t)|$ with $a/b > 1$, i.e. t has many more private neighbors than s.

4 Facets of the formulation

In this section we determine which constraints of (2)–(5) define facets of $P^{a,b}(G)$. Following Coll et al. [4], the proofs use pairs of points of the face instead of explicit sets of affinely independent points.

4.1 Preliminaries

For $(u,v) \in \overline{E}$ let $W_{uv} = P_{uv} \cup Q_{uv} = N(u) \triangle N(v)$; note that $W_{uv} = W_{vu}$, $P_{vu} = Q_{uv}$, and $u \in Q_{uv}$, $v \in P_{uv}$, so $|W_{uv}| \geq 2$. For $x \in \{0,1\}^V$, the *defect* of x at (u,v) is

$$y_{uv}(x) = x(P_{uv}) + |Q_{uv}| - x(Q_{uv}) = |\{w \in W_{uv} : x(w) \neq \chi^{Q_{uv}}(w)\}|,$$

i.e. the Hamming distance on W_{uv} between x and the characteristic vector of Q_{uv}, and we let

$$\kappa_{uv} = \left\lceil \frac{b\,|Q_{uv}| - a\,|P_{uv}|}{\delta} \right\rceil - 1.$$

Lemma 2. *Let $(u,v) \in \overline{E}$ and $x \in \{0,1\}^V$. Then $c(v) - c(u) = b|Q_{uv}| - a|P_{uv}| - \delta\, y_{uv}(x)$; hence $c(u) < c(v)$ if and only if $y_{uv}(x) \le \kappa_{uv}$. Moreover:*

- *$\kappa_{uv} = \beta_{uv} + |Q_{uv}|$;*
- *$(u,v) \in ST$ if and only if $\kappa_{uv} \ge 0$;*
- *constraint (2) can be written as $y_{uv}(x) + (|W_{uv}| - \kappa_{uv})\, z(u,v) \le |W_{uv}|$, and it is redundant if and only if $\kappa_{uv} \ge |W_{uv}|$.*

Proof. By (1), $c(v) - c(u) = \ell(Q_{uv}) - \ell(P_{uv})$. A vertex $w \in Q_{uv}$ contributes b if $x(w) = 1$ and $a = b - \delta$ otherwise, and a vertex $w \in P_{uv}$ contributes $-a$ if $x(w) = 0$ and $-b = -a - \delta$ otherwise. So every defect lowers $c(v) - c(u)$ by δ from $b|Q_{uv}| - a|P_{uv}|$, which proves the formula. Thus $c(u) < c(v)$ if and only if $y_{uv}(x) < (b|Q_{uv}| - a|P_{uv}|)/\delta$, and since $y_{uv}(x)$ is an integer, if and only if $y_{uv}(x) \le \kappa_{uv}$. Since $d(v) - d(u) = |Q_{uv}| - |P_{uv}|$, we have $\beta_{uv} + |Q_{uv}| = \lceil (a|Q_{uv}| - a|P_{uv}| + \delta|Q_{uv}|)/\delta \rceil - 1 = \kappa_{uv}$. By definition, $(u,v) \in ST$ if and only if $b|Q_{uv}| - a|P_{uv}| > 0$, i.e. $\kappa_{uv} \ge 0$. Finally, $x(P_{uv}) - x(Q_{uv}) = y_{uv}(x) - |Q_{uv}|$ and $M_{uv} = |P_{uv}| - \beta_{uv} = |W_{uv}| - \kappa_{uv}$, so (2) becomes $y_{uv}(x) - |Q_{uv}| + (|W_{uv}| - \kappa_{uv})z(u,v) \le |P_{uv}|$, as claimed; and $M_{uv} \le 0$ if and only if $\kappa_{uv} \ge |W_{uv}|$. □

For example, $\kappa_{uv} = |N(v) \backslash N(u)| - 1$ when $(a,b) = (0,1)$, and $\kappa_{uv} = \lceil |W_{uv}|/2 \rceil - 1$ when $(a,b) = (-1,1)$. In the latter case $\kappa_{uv} = 0$ if and only if $W_{uv} = \{u,v\}$, i.e. u and v are closed twins.

For $x \in \{0,1\}^V$ let $R(x) = \{(u,v) \in \overline{E} : c(u) < c(v)\}$ be the set of arcs *realized* by x. A set $C \subseteq \overline{E}$ is *realizable* if $C \subseteq R(x)$ for some x. Note that $R(x)$ contains at most one of (u,v) and (v,u).

Lemma 3. *The integer points of $P^{a,b}(G)$ are the vectors $(x,z) \in \{0,1\}^V \times \{0,1\}^{\overline{E}}$ such that $z \le \chi^{R(x)}$. In particular, $(x, \mathbf{0}) \in P^{a,b}(G)$ for every x, and decreasing a coordinate of z from 1 to 0 preserves feasibility.*

Proof. This is (6), since (3) holds automatically. □

Lemma 4. *Let $(s,t) \in \overline{E}$, $U \subseteq V$ and $\bar{x} \in \{0,1\}^U$. There is $x \in \{0,1\}^V$ with $x|_U = \bar{x}$ and $(s,t) \in R(x)$ if and only if*

$$|\{w \in U \cap W_{st} : \bar{x}(w) \ne \chi^{Q_{st}}(w)\}| \le \kappa_{st}.$$

In particular, if $\kappa_{st} = 0$, every x that realizes (s,t) coincides with $\chi^{Q_{st}}$ on W_{st}.

Proof. Every extension x of $\bar{x}$ has $y_{st}(x)$ at least the left-hand side, with equality when $x = \chi^{Q_{st}}$ on $W_{st} \backslash U$. Apply Lemma 2. □

By (1), $c(u) - c(v)$ depends only on $x|_{W_{uv}}$, and changing $x(w)$ for one $w \in W_{uv}$ changes it by $\pm\delta$.

The two-point method. We use the following classical characterization (see [6] and [4]).

Theorem 1. *Let P be a polytope with minimal equation system $A^{=}w = b^{=}$, and let $\pi w \le \pi_0$ be a valid inequality defining a face F with $\emptyset \ne F \ne P$. Then F is a facet of P if and only if, for every (λ, λ_0) such that $\lambda w = \lambda_0$ for all $w \in F$, there are u and α with $(\lambda, \lambda_0) = (uA^{=} + \alpha\pi, ub^{=} + \alpha\pi_0)$.*

By Proposition 3, the minimal equation system of $P^{a,b}(G)$ is $z(e) = 0$, $e \in \overline{E} \backslash ST$. So, for a valid inequality $\pi x + \sigma z \le \pi_0$ with face F, $\emptyset \ne F \ne P^{a,b}(G)$, we must show that every equation $\lambda x + \mu z = \lambda_0$ satisfied by all points of F has

$$\lambda = \alpha\pi \qquad \text{and} \qquad \mu_e = \alpha\sigma_e \text{ for all } e \in ST \tag{8}$$

for some $\alpha \in \mathbb{R}$ (the coefficients μ_e, $e \notin ST$, are arbitrary, and $\lambda_0 = \alpha\pi_0$ then follows by evaluating at a point of F). Each pair of points $p, p' \in F$ gives $(\lambda, \mu) \cdot (p - p') = 0$. Typically:

- if $p' = p + e_{x(r)}$ (the points differ only in $x(r)$), then $\lambda_r = 0$;
- if $p' = p + e_{z(f)}$ (the points differ only in $z(f)$), then $\mu_f = 0$.

For instance, Proposition 3 follows in this way: for $\pi = \mathbf{0}$ the pairs $(\mathbf{0}, \mathbf{0}), (e_r, \mathbf{0})$ give $\lambda_r = 0$, and the pairs $(\chi^{Q_{st}}, \mathbf{0}), (\chi^{Q_{st}}, e_{st})$, $(s,t) \in ST$, give $\mu_{st} = 0$, so every equation of $P^{a,b}(G)$ is a combination of $z(e) = 0$, $e \in \overline{E} \backslash ST$. Conversely, if all points of F satisfy an equation $\lambda x + \mu z = \lambda_0$ that does not satisfy (8), then F is not a facet. We use this in the form: if $z(f) = 0$ on F for some $f \in ST$ with $\sigma_f = 0$, or if F satisfies an equation whose x-part is not a multiple of π, then F is not a facet.

4.2 Bounds

Theorem 2. *(a) For every $(u,v) \in ST$, the inequality $z(u,v) \geq 0$ defines a facet of $P^{a,b}(G)$.*

(b) For $r \in V$, the inequality $x(r) \geq 0$ defines a facet if and only if $\kappa_{st} \geq 1$ for every $(s,t) \in ST$ with $r \in Q_{st}$.

(c) For $r \in V$, the inequality $x(r) \leq 1$ defines a facet if and only if $\kappa_{st} \geq 1$ for every $(s,t) \in ST$ with $r \in P_{st}$.

Proof. (a) Let $F = \{(x,z) \in P^{a,b}(G) : z(u,v) = 0\}$. Then $(\mathbf{0}, \mathbf{0}) \in F$, and $(\chi^{Q_{uv}}, e_{uv}) \in P^{a,b}(G) \backslash F$ because $y_{uv}(\chi^{Q_{uv}}) = 0 \leq \kappa_{uv}$. Let $\lambda x + \mu z = \lambda_0$ hold on F.

- For $r \in V$: the points $(\mathbf{0}, \mathbf{0})$ and $(e_r, \mathbf{0})$ belong to F, so $\lambda_r = 0$.
- For $(s,t) \in ST \backslash \{(u,v)\}$: the points $(\chi^{Q_{st}}, \mathbf{0})$ and $(\chi^{Q_{st}}, e_{st})$ belong to F, so $\mu_{st} = 0$.

Hence (8) holds with $\alpha = -\mu_{uv}$.

(b) Let $F = \{(x,z) \in P^{a,b}(G) : x(r) = 0\}$. Then $(\mathbf{0}, \mathbf{0}) \in F$ and $(e_r, \mathbf{0}) \notin F$. Assume the condition and let $\lambda x + \mu z = \lambda_0$ hold on F.

- For $r' \neq r$: $(\mathbf{0}, \mathbf{0}), (e_{r'}, \mathbf{0}) \in F$, so $\lambda_{r'} = 0$.
- For $(s,t) \in ST$: apply Lemma 4 with $U = \{r\}$ and $\bar{x}(r) = 0$. The left-hand side is 1 if $r \in Q_{st}$ and 0 otherwise; in both cases it is at most κ_{st}. So some x with $x(r) = 0$ realizes (s,t), and $(x, \mathbf{0}), (x, e_{st}) \in F$ give $\mu_{st} = 0$.

Hence (8) holds with $\alpha = -\lambda_r$. Conversely, if $r \in Q_{st}$ and $\kappa_{st} = 0$ for some $(s,t) \in ST$, then by Lemma 4 no x with $x(r) = 0$ realizes (s,t), so $z(s,t) = 0$ on F, and F is not a facet.

(c) The same argument with $\bar{x}(r) = 1$ (the left-hand side is 1 if and only if $r \in P_{st}$), the points $(e_r, \mathbf{0})$ and $(e_r + e_{r'}, \mathbf{0})$, and $(\mathbf{0}, \mathbf{0}) \notin F$. □

For $(s,t) \notin ST$ the inequality $z(s,t) \geq 0$ is an implicit equation. The conditions of (b) and (c) fail exactly when fixing the label of r forbids some arc.

Corollary 2. *(a) If $(a,b) = (-1,1)$, then $x(r) \geq 0$ and $x(r) \leq 1$ define facets if and only if r has no closed twin.*

(b) If $(a,b) = (0,1)$, then $x(r) \geq 0$ defines a facet if and only if every neighbor t of r has a neighbor outside $N[r]$, and $x(r) \leq 1$ defines a facet if and only if there is no arc (s,t) with $s \in N(r)$, $r \notin N(t)$ and $N(t) \subseteq N[s]$ (possibly $t = r$).

Proof. (a) Here $\kappa_{st} = 0$ if and only if s and t are closed twins, and then $Q_{st} = \{s\}$, $P_{st} = \{t\}$. (b) Here $ST = \overline{E}$ and $\kappa_{st} = 0$ if and only if $Q_{st} = \{s\}$, i.e. $N(t) \subseteq N[s]$. Since $s \in Q_{st}$, $r \in Q_{st} = \{s\}$ means $r = s$; and $r \in P_{st}$ means $s \in N(r)$ and $r \notin N(t)$. □

4.3 The edge constraints (3)

At least one of the arcs (u, v) and (v, u) belongs to ST: otherwise adding $b|Q_{uv}| \le a|P_{uv}|$ and $b|P_{uv}| \le a|Q_{uv}|$ would give $b|W_{uv}| \le a|W_{uv}|$. So we may assume $(u, v) \in ST$.

Theorem 3. *Let $uv \in E$ with $(u, v) \in ST$. The inequality $z(u, v) + z(v, u) \le 1$ defines a facet of $P^{a,b}(G)$ if and only if*

(A) $\kappa_{uv} \ge 1$ or $\kappa_{vu} \ge 1$, and

(B) there is no arc $(s, t) \in ST \backslash \{(u, v), (v, u)\}$ with $\kappa_{st} = 0$, $W_{uv} \subseteq W_{st}$ and $c(u) = c(v)$ for the labeling $\chi^{Q_{st}}$.

Proof. Let F be the face. Then $(\mathbf{0}, \mathbf{0}) \notin F$ and $(\chi^{Q_{uv}}, e_{uv}) \in F$. Assume (A) and (B) and let $\lambda x + \mu z = \lambda_0$ hold on F. If $\kappa_{uv} \ge 1$, let $e^* = (u, v)$; otherwise $\kappa_{vu} \ge 1$ (so $(v, u) \in ST$) and let $e^* = (v, u)$. Let $x^* = \chi^{Q_{e^*}}$.

- $\lambda_r = 0$ *for* $r \in V$. Let $x' = x^* + (1 - 2x^*(r))\, e_r$ be x^* with the label of r changed. Then $y_{e^*}(x') \le 1 \le \kappa_{e^*}$ (it is 1 if $r \in W_{uv}$ and 0 otherwise), so (x^*, e_{e^*}) and (x', e_{e^*}) belong to F, and $\lambda_r = 0$.

- $\mu_{st} = 0$ *for* $(s, t) \in ST \backslash \{(u, v), (v, u)\}$. It is enough to find x that realizes (s, t) and has $c(u) \ne c(v)$: then $R(x)$ contains $e = (u, v)$ or $e = (v, u)$, and the points (x, e_e) and $(x, e_e + e_{st})$ belong to F. Let $x_0 = \chi^{Q_{st}}$, which realizes (s, t). If $c(u) \ne c(v)$ at x_0 we are done. Otherwise, if some $w \in W_{uv}$ is not in W_{st}, changing the label of w keeps $y_{st} = 0$ and changes $c(u) - c(v)$ by $\pm\delta$. If $W_{uv} \subseteq W_{st}$, then $\kappa_{st} \ge 1$ by (B), and changing the label of any $w \in W_{uv}$ gives $y_{st} = 1 \le \kappa_{st}$ and again changes $c(u) - c(v)$ by $\pm\delta$.

- μ_{uv} *and* μ_{vu}. The point $(\chi^{Q_{uv}}, e_{uv}) \in F$ gives $\mu_{uv} = \lambda_0$, and, if $(v, u) \in ST$, the point $(\chi^{P_{uv}}, e_{vu}) \in F$ gives $\mu_{vu} = \lambda_0$.

Hence (8) holds with $\alpha = \lambda_0$.

Conversely, suppose that (A) fails. Then every point of F realizes (u, v) with $y_{uv} = 0$ or (v, u) with $y_{vu} = 0$, so its x coincides on W_{uv} with $\chi^{Q_{uv}}$ or with $\chi^{P_{uv}}$. As $u \in Q_{uv}$ and $v \in P_{uv}$, all points of F satisfy $x(u) + x(v) = 1$, and F is not a facet. Suppose that (B) fails for (s, t). By Lemma 4, every x realizing (s, t) coincides with $\chi^{Q_{st}}$ on $W_{st} \supseteq W_{uv}$; since $c(u) - c(v)$ depends only on $x|_{W_{uv}}$, every such x violates uv. Hence no point of F has $z(s, t) = 1$, and F is not a facet. □

Corollary 3. *Let $(u, v) \in ST$. The inequality $z(u, v) \le 1$ defines a facet of $P^{a,b}(G)$ if and only if $(v, u) \notin ST$, $\kappa_{uv} \ge 1$ and condition (B) of Theorem 3 holds.*

Proof. If $(v, u) \in ST$, every point of the face has $z(v, u) = 0$, so it is not a facet. If $(v, u) \notin ST$, then $z(v, u) = 0$ is an equation of $P^{a,b}(G)$, the faces of $z(u, v) \le 1$ and of (3) coincide, and (A) reads $\kappa_{uv} \ge 1$. □

Corollary 4. *If $(a, b) = (-1, 1)$, then (3) defines a facet if and only if u and v are not closed twins.*

Proof. Condition (A) says that u and v are not closed twins. If $\kappa_{st} = 0$, then $W_{st} = \{s, t\}$, so $W_{uv} \subseteq W_{st}$ forces $\{u, v\} = \{s, t\}$, and (B) holds trivially. □

4.4 The constraints (2)

By Lemma 2, (2) is a proper constraint only when $0 \le \kappa_{uv} \le |W_{uv}| - 1$. Its facets are characterized in Corollary 5 below, as the case $S = W_{uv}$ of a larger family of valid inequalities.

5 New valid inequalities

5.1 Cover inequalities

Let $(u,v) \in ST$ and $S \subseteq W_{uv}$, and write $y^S_{uv}(x) = x(S \cap P_{uv}) + |S \cap Q_{uv}| - x(S \cap Q_{uv})$ for the number of defects inside S. If $|S| \geq \kappa_{uv} + 1$, the *cover inequality* is

$$y^S_{uv}(x) + (|S| - \kappa_{uv})\, z(u,v) \leq |S|, \qquad \text{i.e.} \qquad x(S \cap P_{uv}) - x(S \cap Q_{uv}) + (|S| - \kappa_{uv})\, z(u,v) \leq |S \cap P_{uv}|. \tag{9}$$

It is valid: if $z(u,v) = 0$ it says $y^S_{uv}(x) \leq |S|$, and if $z(u,v) = 1$ then $y^S_{uv}(x) \leq y_{uv}(x) \leq \kappa_{uv}$ by Lemma 2. For $S = W_{uv}$ it is (2), by Lemma 2. The family contains every strengthening of (2) that involves only x and $z(u,v)$:

Proposition 4. *Let $(u,v) \in \overline{E}$ with $0 \leq \kappa_{uv} < |W_{uv}|$, and let $Y = \{(x,z) \in \{0,1\}^{W_{uv}} \times \{0,1\} : z = 1 \Rightarrow y_{uv}(x) \leq \kappa_{uv}\}$. Then $\mathrm{conv}(Y)$ is described by the bounds $0 \leq x \leq 1$, $0 \leq z \leq 1$ and the cover inequalities (9) for all $S \subseteq W_{uv}$ with $|S| \geq \kappa_{uv} + 1$.*

Proof. Complementing the variables of Q_{uv} we may assume $Q_{uv} = \emptyset$, so $y_{uv}(x) = x(W)$ with $W = W_{uv}$, and we write $\kappa = \kappa_{uv}$. Then $\mathrm{conv}(Y) = \mathrm{conv}(Y_0 \cup Y_1)$ with $Y_0 = [0,1]^W \times \{0\}$ and $Y_1 = \{x \in [0,1]^W : x(W) \leq \kappa\} \times \{1\}$ (both are integral polytopes). Let (x,z) satisfy the bounds and all covers. If $z \in \{0,1\}$ the claim is clear (for $z = 1$, take $S = W$). Otherwise, $(x,z) = (1-z)(x^0, 0) + z(x^1, 1)$ with $x^0, x^1 \in [0,1]^W$ if and only if $\max(0, (x_w - 1 + z)/z) \leq x^1_w \leq \min(1, x_w/z)$ for all w; these intervals are nonempty. Such an x^1 with $x^1(W) \leq \kappa$ exists if and only if $\sum_w \max(0, x_w + z - 1) \leq \kappa z$. The left-hand side equals $\max_{S \subseteq W}(x(S) + |S|z - |S|)$, so the condition says $x(S) + (|S| - \kappa)z \leq |S|$ for all S, which holds by the covers (and by the bounds when $|S| \leq \kappa$). □

To state when (9) defines a facet of $P^{a,b}(G)$, for arcs $e = (u,v)$ and $f = (s,t)$ let

$$A_{ef} = (P_{uv} \cap P_{st}) \cup (Q_{uv} \cap Q_{st}), \qquad D_{ef} = (P_{uv} \cap Q_{st}) \cup (Q_{uv} \cap P_{st}),$$

the vertices of $W_{uv} \cap W_{st}$ on the same side and on opposite sides. So a defect for e at a vertex of A_{ef} is also a defect for f, and a defect for e at a vertex of D_{ef} is not.

Theorem 4. *Let $e = (u,v) \in ST$, $\kappa = \kappa_{uv}$ and $S \subseteq W_{uv}$ with $|S| \geq \kappa + 1$. The cover inequality (9) defines a facet of $P^{a,b}(G)$ if and only if*

(A_S) $\kappa \geq 1$ or $|S| = 1$, and

(B_S) for every $f = (s,t) \in ST \backslash \{e\}$, either $|S \cap A_{ef}| \leq \kappa_{st}$, or $f \neq (v,u)$ and

$$|D_{ef}| - \min(\kappa, |S \cap D_{ef}|) + \max(0, \kappa - |S \cap D_{ef}| - |S \backslash W_{st}|) \leq \kappa_{st}.$$

Proof. Let F be the face. By Lemma 3, an integer point $(x,z) \in P^{a,b}(G)$ lies in F if and only if it is of one of two types:

- *type 0*: $z(u,v) = 0$ and $y^S_{uv}(x) = |S|$, i.e. $x = \chi^{P_{uv}}$ on S;
- *type 1*: $z(u,v) = 1$ and $y^S_{uv}(x) = \kappa$. Since $\kappa = y^S_{uv}(x) \leq y_{uv}(x) \leq \kappa$, this means that $x = \chi^{Q_{uv} \triangle T}$ on W_{uv} for some $T \subseteq S$ with $|T| = \kappa$ (the defects are exactly at T).

In both cases the remaining labels and the remaining coordinates of z are arbitrary, subject to $z \le \chi^{R(x)}$. The point $(\chi^{Q_{uv}}, \mathbf{0})$ has $y^S_{uv} = 0 < |S|$, so $F \ne P^{a,b}(G)$; and $F \ne \emptyset$.

The existence of points with $z(f) = 1$. Let $f = (s,t) \in ST\backslash\{e\}$. By Lemma 4 with $U = S$, there is a point of type 0 realizing f if and only if $|S \cap A_{ef}| \le \kappa_{st}$ (on S the labeling is $\chi^{P_{uv}}$, which differs from $\chi^{Q_{st}}$ exactly on $S \cap A_{ef}$). If $f = (v,u)$ there is no point of type 1 with $z(f) = 1$, and $A_{ef} = \emptyset$. Otherwise, by Lemma 4 with $U = W_{uv}$, there is a point of type 1 with defect set T realizing f if and only if $|D_{ef}\backslash T| + |T \cap A_{ef}| \le \kappa_{st}$. To minimize the left-hand side over $T \subseteq S$ with $|T| = \kappa$, take first the elements of $S \cap D_{ef}$, then those of $S\backslash W_{st}$, and only then those of $S \cap A_{ef}$; the minimum is the left-hand side of the inequality in (B_S). Therefore (B_S) holds if and only if for every $f \in ST\backslash\{e\}$ some point of F has $z(f) = 1$.

Sufficiency. Assume (A_S) and (B_S) and let $\lambda x + \mu z = \lambda_0$ hold on F. For $w \in W_{uv}$ let $\tilde\lambda_w = \lambda_w$ if $w \in P_{uv}$ and $\tilde\lambda_w = -\lambda_w$ if $w \in Q_{uv}$; so creating a defect at w changes λx by $\tilde\lambda_w$.

- $\lambda_r = 0$ *for* $r \notin S$. Let x be of type 0. Then $(x, \mathbf{0})$ and $(x', \mathbf{0})$ belong to F, where x' is x with the label of r changed.
- $\mu_f = 0$ *for* $f \in ST\backslash\{e\}$. Take $(x,z) \in F$ with $z(f) = 1$. Then $(x, z - e_f) \in F$.
- $\tilde\lambda$ *is constant on* S. There is nothing to prove if $|S| = 1$. Otherwise $1 \le \kappa \le |S| - 1$ by (A_S). For $r, r' \in S$ choose $T \subseteq S$ with $|T| = \kappa$, $r \in T$ and $r' \notin T$, and let $T' = (T\backslash\{r\}) \cup \{r'\}$. The two points of type 1 with defect sets T and T', both $\mathbf{0}$ outside W_{uv} and with $z = e_{uv}$, differ only in the labels of r and r', so $\tilde\lambda_r - \tilde\lambda_{r'} = 0$. Let θ be the common value.
- $\mu_{uv} = \theta(|S| - \kappa)$. Fix $T \subseteq S$ with $|T| = \kappa$. Let $x^1 = \chi^{Q_{uv} \triangle T}$ on W_{uv} and $\mathbf{0}$ elsewhere, and let x^0 be x^1 with defects also on $S\backslash T$. Then $(x^0, \mathbf{0})$ is of type 0 and (x^1, e_{uv}) is of type 1, so $\lambda x^0 = \lambda_0 = \lambda x^1 + \mu_{uv}$, and $\lambda x^0 - \lambda x^1 = \sum_{w \in S\backslash T} \tilde\lambda_w = \theta(|S| - \kappa)$.

Since the coefficient of $x(w)$ in (9) is 1 for $w \in S \cap P_{uv}$ and -1 for $w \in S \cap Q_{uv}$, (8) holds with $\alpha = \theta$.

Necessity. If (B_S) fails for f, then $z(f) = 0$ on F, and F is not a facet. If (A_S) fails, then $\kappa = 0$ and $|S| \ge 2$, so the points of type 1 have no defects on W_{uv}, and those of type 0 have defects at every vertex of S. For distinct $r, r' \in S$, every point of F thus has a defect at both or at neither, i.e. it satisfies $\pm x(r) \mp x(r') =$ constant (with the signs given by the sides of r and r'). This equation has no z-terms and a nonzero x-part that is not a multiple of the x-part of (9), so F is not a facet. □

Corollary 5. *Let $(u,v) \in ST$ with $\kappa_{uv} \le |W_{uv}| - 1$. Constraint (2) defines a facet of $P^{a,b}(G)$ if and only if $\kappa_{uv} \ge 1$ and (B_S) holds for $S = W_{uv}$. If $\kappa_{uv} = 0$, (2) is the sum of the $|W_{uv}|$ cover inequalities with $|S| = 1$:*

$$z(u,v) \le x(w) \ \ (w \in Q_{uv}), \qquad z(u,v) \le 1 - x(w) \ \ (w \in P_{uv}).$$

For example, if u and v are closed twins, then $W_{uv} = \{u,v\}$, $c(u) - c(v) = \ell(v) - \ell(u)$ and $\kappa_{uv} = \lceil (b-a)/\delta \rceil - 1 = 0$ for every $a < b$. The covers are $z(u,v) \le x(u)$ and $z(u,v) \le 1 - x(v)$: $c(u) < c(v)$ forces $\ell(u) = b$ and $\ell(v) = a$. For $(a,b) = (-1,1)$ these are the only arcs with $\kappa_{uv} = 0$; for $(a,b) = (0,1)$, $\kappa_{uv} = 0$ if and only if $N(v) \subseteq N[u]$.

The covers with $S \ne W_{uv}$ really cut off fractional points of the linear relaxation. If $\kappa_{uv} = 1$ and $|W_{uv}| = 3$, then the point with defects on two vertices S of W_{uv}, no defect on the third one, and $z(u,v) = 1/2$ satisfies (2) ($2 + 2 \cdot \frac{1}{2} = 3$) but violates the cover $y^S_{uv}(x) + z(u,v) \le 2$.

5.2 Conflict inequalities

A *conflict clique* is a nonempty set $C \subseteq ST$ such that no two arcs of C are realizable together, i.e. $|C \cap R(x)| \leq 1$ for every x. Then

$$\sum_{e \in C} z(e) \leq 1 \tag{10}$$

is valid. Constraint (3) is the case $C = \{(u,v),(v,u)\}$ and $z(u,v) \leq 1$ is the case $|C| = 1$.

Theorem 5. *Let C be a conflict clique. Then (10) defines a facet of $P^{a,b}(G)$ if and only if*

(K1) for every $r \in V$ there is $e \in C$ with $r \notin W_e$ or $\kappa_e \geq 1$, and

(K2) for every $f \in ST \backslash C$ there is $e \in C$ such that $\{e, f\}$ is realizable.

Proof. Let F be the face; it consists of the integer points with exactly one $z(e) = 1$, $e \in C$. Then $(\mathbf{0}, \mathbf{0}) \notin F$ and $(\chi^{Q_e}, e_e) \in F$ for $e \in C$. Assume (K1), (K2), and let $\lambda x + \mu z = \lambda_0$ hold on F.

- $\lambda_r = 0$. Take $e \in C$ as in (K1) and let x' be χ^{Q_e} with the label of r changed. Then $y_e(x') \leq \kappa_e$, so (χ^{Q_e}, e_e) and (x', e_e) belong to F.
- $\mu_f = 0$ *for* $f \in ST \backslash C$. Take e as in (K2) and x realizing $\{e, f\}$; then (x, e_e) and $(x, e_e + e_f)$ belong to F.
- $\mu_e = \lambda_0$ *for* $e \in C$, from the point (χ^{Q_e}, e_e).

So (8) holds with $\alpha = \lambda_0$. Conversely, if (K2) fails for f, then $z(f) = 0$ on F. If (K1) fails for r, then for every $e \in C$ we have $r \in W_e$ and $\kappa_e = 0$, so by Lemma 4 every x realizing e has $x(r) = \chi^{Q_e}(r)$. Hence F satisfies the equation $x(r) = \sum_{e \in C} \chi^{Q_e}(r)\, z(e)$, whose x-part is not zero. In both cases F is not a facet. □

For $C = \{(u,v),(v,u)\}$, (K1) is condition (A) of Theorem 3 and (K2) is condition (B), in the explicit form proved there. More generally, if C is a nonrealizable set such that $C \backslash \{e\}$ is realizable for every $e \in C$, then the *circuit inequality*

$$\sum_{e \in C} z(e) \leq |C| - 1 \tag{11}$$

is valid.

Theorem 6. *Let C be as above, with $|C| \geq 2$. Inequality (11) defines a facet of $P^{a,b}(G)$ if*

(C1) for every $r \in V$ there are $e \in C$ and x such that both x and x with the label of r changed realize $C \backslash \{e\}$, and

(C2) for every $f \in ST \backslash C$ there is $e \in C$ such that $(C \backslash \{e\}) \cup \{f\}$ is realizable.

Condition (C2) is also necessary.

Proof. The face F consists of the integer points with $z(e) = 1$ for all but one $e \in C$. (C1) gives $\lambda_r = 0$ and (C2) gives $\mu_f = 0$, exactly as in Theorem 5. For $e \in C$, a point $(x, \chi^{C \backslash \{e\}}) \in F$ gives $\mu(C) - \mu_e = \lambda_0$, so all μ_e, $e \in C$, are equal to some θ, and (8) holds with $\alpha = \theta$. If (C2) fails for f, then $z(f) = 0$ on F. □

For $|C| = 2$ the circuit and clique inequalities coincide, and (C1) is (K1). For $|C| \geq 3$, (C1) is not necessary: on small random graphs, between 10% and 20% of the facet-defining circuit inequalities with $|C| = 3$ violate it.

5.3 Sources of conflicts

The following lemma gives a simple sufficient condition for an arc set to be nonrealizable. For vertices p, q (adjacent or not), the same argument as in (7) shows that

$$\max_x \big(c(p) - c(q)\big) = b\,|N(p)\backslash N(q)| - a\,|N(q)\backslash N(p)|. \tag{12}$$

We write $p \preceq q$ (q *dominates* p) if this maximum is at most 0, i.e. $c(p) \le c(q)$ for every labeling. For adjacent s, t, $(s,t) \notin ST$ if and only if $t \preceq s$. If $a \ge 0$, then $N(p) \subseteq N(q)$ implies $p \preceq q$. If $a < 0$ this is false in general, but false twins ($N(p) = N(q)$) always satisfy $p \preceq q$ and $q \preceq p$.

Proposition 5. *Let $(s_1,t_1),\dots,(s_k,t_k) \in \overline{E}$, $k \ge 1$, with indices modulo k. Let $h_i = \max_x(c(t_i) - c(s_{i+1}))$, given by (12) (with $h_i = 0$ if $t_i = s_{i+1}$), and let g_i be the least positive integer congruent to $a\,(d(t_i) - d(s_i))$ modulo δ. If $\sum_i g_i > \sum_i h_i$, then $\{(s_1,t_1),\dots,(s_k,t_k)\}$ is not realizable, and $\sum_{i=1}^k z(s_i,t_i) \le k-1$ is valid.*

Proof. Since $c(w) = a\,d(w) + \delta\,x(N(w))$, the difference $c(t_i) - c(s_i)$ is congruent to $a(d(t_i) - d(s_i))$ modulo δ, so it is at least g_i whenever it is positive. If some x realized all the arcs, the telescoping identity $0 = \sum_i \big(c(t_i) - c(s_i)\big) + \sum_i \big(c(s_{i+1}) - c(t_i)\big) \ge \sum_i g_i - \sum_i h_i$ would give a contradiction. □

Since $g_i \ge 1$, the hypothesis holds whenever $t_i \preceq s_{i+1}$ for all i (*dominance cycles*). Special cases:

- $k = 2$, $(s_2,t_2) = (t_1,s_1)$: constraint (3).
- Directed cycles $t_i = s_{i+1}$: the cycle inequalities $\sum_{i=1}^k z(v_i,v_{i+1}) \le k-1$ for a closed walk $v_1v_2\cdots v_kv_1$.
- *2-chains*: $(u,w),(w,v)$ with $u \ne v$ and $v \preceq u$, which give $z(u,w) + z(w,v) \le 1$. If $a \ge 0$ this applies whenever $N(v) \subseteq N(u)$. For $a < 0$ the inclusion is not enough, but $N(u) = N(v)$ is.
- The induced P_4 $u'vv'u$ with $u \preceq v$ and $u' \preceq v'$: arcs (v',u) and (v,u'), giving $z(v',u) + z(v,u') \le 1$.
- *Gaps*: for $(a,b) = (0,1)$ all $g_i = 1$, so a 2-chain $(u,w),(w,v)$ is a conflict as soon as $|N(v)\backslash N(u)| \le 1$. For example, in the path 0 1 2 3, $z(3,2) + z(2,1) \le 1$ is a facet, although $1 \not\preceq 3$.
- *Closed twins*: if u,v,w are pairwise closed twins, then $c(p) = \ell(N[u]) - \ell(p)$ for $p \in \{u,v,w\}$ takes only two values, so any two of the arcs $(u,v),(v,w),(w,u)$ form a nonrealizable 2-chain, and $z(u,v) + z(v,w) + z(w,u) \le 1$ is a conflict clique of size 3.

In all these cases the facet conditions are those of Theorems 5 and 6. For a 2-chain $C = \{(u,w),(w,v)\}$, (K1) says that $\max(\kappa_{uw},\kappa_{wv}) \ge 1$ whenever $W_{uw} \cap W_{wv} \ne \emptyset$. For instance, in K_3 both directed triangles define facets.

5.4 Conditional conflicts and the linear relaxation

Fixing some labels creates more conflicts. If $C \subseteq \overline{E}$ and $U_0, U_1 \subseteq V$ are disjoint sets such that C is not realized by any x with $x|_{U_0} = 0$ and $x|_{U_1} = 1$, then

$$\sum_{e \in C} z(e) \le |C| - 1 + x(U_0) + |U_1| - x(U_1) \tag{13}$$

is valid, since the right-hand side is at least $|C|$ outside that subcube. The covers with $|S| = \kappa_{uv} + 1$ are the case $|C| = 1$. In the small graphs we enumerated, many facets have this form. For example, for the

paw (a triangle 0 1 2 with a pendant vertex 3 at 2) and $(a, b) = (-1, 1)$, the inequality $z(1,2) + z(2,0) \leq 1 + x(1)$ defines a facet. Even richer facets exist, e.g. $x(0) - x(2) + z(1,0) + z(2,0) + z(2,1) \leq 1$ for K_3. Characterizing the facets of (13) is left open.

The following observation explains why the cliques matter most for the lower bound.

Proposition 6. *Assume $ST = \overline{E}$ (e.g. $a \leq 0 \leq b$). The point $x = \frac{1}{2}\mathbf{1}$, $z = \frac{1}{2}\mathbf{1}$ satisfies (2), (3), all bounds, all cover inequalities, all circuit inequalities, and all inequalities (13) with $|C| \geq 2$ or $U_0 \cup U_1 \neq \emptyset$. Its objective value is 0.*

Proof. For a cover, $\frac{|S|}{2} + \frac{|S| - \kappa_{uv}}{2} \leq |S|$ since $\kappa_{uv} \geq 0$; (2) is the case $S = W_{uv}$. For a circuit, $\frac{|C|}{2} \leq |C| - 1$ when $|C| \geq 2$. For (13), $\frac{|C|}{2} \leq |C| - 1 + \frac{|U_0| + |U_1|}{2}$ under the stated hypothesis. Finally, $|E| - \sum_e z(e) = |E| - \frac{1}{2} \cdot 2|E| = 0$. □

Therefore, when $ST = \overline{E}$, the linear relaxation, even strengthened with all covers, all 2-conflicts and all circuits, always gives the trivial bound 0. Only conflict cliques with $|C| \geq 3$ (and inequalities with mixed x- and z-terms) cut off this point, since $\frac{|C|}{2} > 1$. On small examples the cliques close the gap: with all maximal conflict cliques the bound is $\eta_v^{a,b}(G)$ for K_3 (value 1), K_4 (2) and W_3 (2), for $(a, b) \in \{(0,1), (-1,1), (1,2)\}$. For the net and the bowtie with $(a, b) = (-1, 1)$, and for C_5, the bound stays 0 while $\eta_v^{a,b}(G) = 1$: there the obstruction is global, as the Nullstellensatz certificate of Section 7 shows for the net.

6 Symmetries

A *symmetry* is an affine bijection of $\mathbb{R}^V \times \mathbb{R}^{\overline{E}}$ that maps the integer points of $P^{a,b}(G)$ onto themselves and preserves the objective function $|E| - \sum_e z(e)$.

Proposition 7. *(a) Every automorphism π of G induces the symmetry $(x, z) \mapsto (x \circ \pi^{-1}, z \circ \pi^{-1})$, where $(z \circ \pi^{-1})(u, v) = z(\pi^{-1}(u), \pi^{-1}(v))$.*

(b) If $a + b = 0$, or if G is regular, then $\rho(x, z) = (\mathbf{1} - x, z^T)$, where $z^T(u, v) = z(v, u)$, is a symmetry.

(c) If u and u' are false twins, exchanging $x(u)$ and $x(u')$ and keeping z is a symmetry.

Proof. (a) Colors are carried along: $c_{x \circ \pi^{-1}}(\pi(v)) = c_x(v)$, so $R(x \circ \pi^{-1}) = \pi(R(x))$. Apply Lemma 3. (b) $\mathbf{1} - x$ encodes $\ell' = (a + b) - \ell$, with colors $c'(v) = (a + b)\, d(v) - c(v)$. If $(a + b)(d(u) - d(v)) = 0$ for every edge uv, then $c'(u) < c'(v)$ if and only if $c(v) < c(u)$, so $R(\mathbf{1} - x) = R(x)^T$. (c) $N(w)$ contains both or neither of u, u' for every w, so all colors are unchanged, and so is $R(x)$. □

For $b = -a$ the map ρ is $\ell \mapsto -\ell$; in general it is the identity $\eta_v^{a,b} = \eta_v^{-b,-a}$ followed by the shift $\ell \mapsto \ell + a + b$, which preserves the order of the colors only along edges whose endpoints have equal degrees.

Symmetry-breaking inequalities. Let Γ be the group generated by the symmetries of Proposition 7(a), and by ρ when it applies. Fix an order $v_1, \ldots, v_n$ of V and compare labelings lexicographically in this order. Each Γ-orbit of integer points contains a point whose x is lexicographically maximal in the orbit (a *lex-leader*). As the objective is Γ-invariant, adding inequalities satisfied by all lex-leaders does not change the optimal value. For $i \geq 1$, let Aut_i be the subgroup of automorphisms fixing $v_1, \ldots, v_{i-1}$ (so $\mathrm{Aut}_1 = \mathrm{Aut}(G)$).

Theorem 7. *Every lex-leader satisfies*

$$x(v_i) \geq x(w) \qquad \text{for all } i \text{ and all } w \text{ in the orbit of } v_i \text{ under } \mathrm{Aut}_i. \tag{14}$$

If ρ is a symmetry, every lex-leader also satisfies

$$x(v_1) = 1 \qquad \text{and} \qquad x(v_2) + x(\pi(v_1)) + x(\pi(v_2)) \geq 1 \quad \text{for all } \pi \in \mathrm{Aut}(G). \tag{15}$$

Therefore $\min\{|E| - \sum_e z(e) : (x,z) \in P^{a,b}(G) \cap \mathbb{Z}^{V \cup \overline{E}}$ *satisfies (14) and (15)*$\} = \eta_v^{a,b}(G)$.

Proof. Let (x,z) be a lex-leader. For $\pi \in \mathrm{Aut}_i$, the labeling $x \circ \pi^{-1}$ agrees with x on $v_1, \ldots, v_{i-1}$, so $x \geq_{\mathrm{lex}} x \circ \pi^{-1}$ gives $x(v_i) \geq x(\pi^{-1}(v_i))$, and $\pi^{-1}(v_i)$ runs over the orbit. If ρ is a symmetry, $x \geq_{\mathrm{lex}} \mathbf{1} - x$ forces $x(v_1) = 1$. For $\pi \in \mathrm{Aut}(G)$, compare x with $y = \mathbf{1} - x \circ \pi^{-1}$, the image under ρ composed with the symmetry induced by π. If $x(\pi^{-1}(v_1)) = 0$ then $y(v_1) = 1 = x(v_1)$, so $x(v_2) \geq y(v_2) = 1 - x(\pi^{-1}(v_2))$. In all cases $x(v_2) + x(\pi^{-1}(v_1)) + x(\pi^{-1}(v_2)) \geq 1$, and π^{-1} runs over $\mathrm{Aut}(G)$. □

Twins are the main source of automorphisms. If $\{c_1, \ldots, c_m\}$ is a class of pairwise (false or closed) twins listed in this order, every transposition of the class is an automorphism, and (14) contains the chain

$$x(c_1) \geq x(c_2) \geq \cdots \geq x(c_m).$$

For a false-twin class this merely chooses which vertices of the class carry the label b; only their number matters, since the class members have the same neighbors. For a closed-twin class, the chain combines with the covers $z(c_i, c_j) \leq x(c_i)$ and $z(c_i, c_j) \leq 1 - x(c_j)$ (Corollary 5): for $i > j$ they force $z(c_i, c_j) = 0$, since $x(c_i) = 1$ implies $x(c_j) = 1$. Choosing v_1 in a large orbit makes (14) strongest. For instance, for W_k with $v_1 \in A$, the orbit of v_1 is $A \cup B$.

Optimality-based inequalities. The symmetry-breaking inequalities can be combined with the following equations and inequalities, which hold for *every* optimal solution (they are not valid for $P^{a,b}(G)$).

Proposition 8. *Every optimal integer solution (x,z) satisfies $z = \chi^{R(x)}$. Consequently, if $v \preceq u$ and $w \in N(u) \cap N(v)$, then $z(v,w) \geq z(u,w)$ and $z(w,u) \geq z(w,v)$. In particular, for false twins u, u' and $w \in N(u)$: $z(u,w) = z(u',w)$ and $z(w,u) = z(w,u')$.*

Proof. By Lemma 3, raising $z(e)$ to 1 for a realized arc e keeps feasibility and lowers the objective, so $z = \chi^{R(x)}$ at an optimum. If $c(u) < c(w)$, then $c(v) \leq c(u) < c(w)$, and if $c(w) < c(v)$, then $c(w) < c(v) \leq c(u)$. □

Since some lex-leader is optimal, the problem keeps its optimal value when (14), (15) and the constraints of Proposition 8 are all added. In a sample of 117 random instances with at most 7 vertices (most of them with $b = -a$, so that ρ applies), (14) and (15) remove about 63% of the integer points. Added to the formulation alone, they do not raise the bound of the linear relaxation in our examples (it stays 0). Together with the covers and the 2-element conflict cliques they do: the bound becomes $1 = \eta_v^{a,b}(K_3)$ for K_3 and 1.5 for K_4 (where $\eta_v^{a,b}(K_4) = 2$), for each $(a,b) \in \{(0,1), (-1,1), (1,2)\}$.

Interchangeable blocks. Suppose $B_1, \ldots, B_k$ are disjoint vertex sets with bijections $\varphi_j : B_1 \to B_j$ such that, for every permutation σ of $\{1, \ldots, k\}$, the map $\varphi_j(i) \mapsto \varphi_{\sigma(j)}(i)$ (identity outside the blocks) is an automorphism, e.g. k copies of a rooted graph attached to a common vertex. Twin classes are the case $|B_1| = 1$. Read the labels of block j as a word $\omega_j \in \{0,1\}^{B_1}$ and fix a total order $p_1 < p_2 < \cdots < p_{2^m}$ of the words, $m = |B_1|$. On its own (i.e. with an order of V that is not used for (14)), sorting the blocks,

i.e. requiring $\omega_1 \le \omega_2 \le \cdots \le \omega_k$, is symmetry breaking. With pattern variables $\omega(p,j) \in \{0,1\}$, $\sum_p \omega(p,j) = 1$ and $x(\varphi_j(i)) = \sum_{p:p_i=1} \omega(p,j)$, it can be written as

$$\sum_{q \le p} \omega(q, j+1) \le \sum_{q \le p} \omega(q,j) \qquad \text{for all } p \text{ and } j = 1, \dots, k-1.$$

In the cumulative variables $\Omega(p,j) = \sum_{q \le p} \omega(q,j)$ every constraint has the form $\Omega \le \Omega'$ or fixes Ω to 1, so the matrix is totally unimodular and these constraints describe the convex hull of the sorted pattern matrices. The pattern matrix has exactly one 1 per column and the group permutes its columns, so it is not a packing or partitioning orbitope in the sense of Kaibel and Pfetsch [5], where each row has at most one 1. Here the order allows repetitions, since two blocks may carry the same word. This is an extended formulation with $2^m k$ variables, so it is useful only for small blocks.

7 A Nullstellensatz certificate for the net

Let $D = (V, A)$ be an orientation of G and consider the additive graph polynomial

$$q_D(x) = \prod_{(i,j) \in A} \big(x(N(j)) - x(N(i))\big) \in \mathbb{Z}[x_v : v \in V].$$

An (a,b)-labeling ℓ is an (a,b)-additive coloring if and only if $q_D(\ell) \neq 0$. Hence $\eta_v^{a,b}(G) > 0$ if and only if q_D vanishes on the grid $\{a,b\}^V$.

Lemma 5 (Grid reduction)**.** *Let $\mathbb{F}$ be a field, $a \neq b$ in $\mathbb{F}$ and $g_v = (x_v - a)(x_v - b)$ for $v \in V$. Every $f \in \mathbb{F}[x]$ can be written as $f = \sum_v h_v g_v + \bar{f}$ with $\bar{f}$ multilinear, and $\bar{f}$ is the unique multilinear polynomial that agrees with f on $\{a,b\}^V$. In particular, f vanishes on $\{a,b\}^V$ if and only if $\bar{f} = 0$, if and only if $f \in \langle g_v : v \in V \rangle$.*

Proof. Replacing repeatedly x_v^2 by $(a+b)x_v - ab$ yields $\bar{f}$, and $f - \bar{f} \in \langle g_v \rangle$ vanishes on the grid. A multilinear polynomial vanishing on $\{a,b\}^V$ is zero: write it as $p_0 + x_v p_1$ with p_0, p_1 free of x_v; then $p_0 + a p_1$ and $p_0 + b p_1$ vanish on the grid of the remaining variables, so by induction $p_0 + a p_1 = p_0 + b p_1 = 0$, and $p_1 = p_0 = 0$ since $a \neq b$. This gives uniqueness, and the last equivalence follows. (This is the grid lemma behind Alon's Combinatorial Nullstellensatz [1].) □

The reduction can be performed with a and b as indeterminates. Indeed, $x^e \equiv \alpha_e x + \beta_e \pmod{(x-a)(x-b)}$, where

$$\alpha_e = \frac{a^e - b^e}{a - b}, \qquad \beta_e = \frac{a\, b^e - b\, a^e}{a - b}$$

belong to $\mathbb{Z}[a,b]$. Replacing every power x_v^e of every monomial of q_D by $\alpha_e x_v + \beta_e$ gives a multilinear polynomial $R_D \in \mathbb{Z}[a,b][x]$ such that, for all $a_0 \neq b_0$, the specialization $R_D(a_0, b_0; x)$ is the reduction of q_D of Lemma 5. Therefore, for integers $a < b$,

$$\eta_v^{a,b}(G) > 0 \iff R_D(a,b;x) = 0 \iff \text{every coefficient of } R_D \text{ vanishes at } (a,b). \tag{16}$$

Since q_D is homogeneous of degree $|E|$ and α_e, β_e are homogeneous of degrees $e-1$ and e, the coefficient of $\prod_{v \in S} x_v$ in R_D is a homogeneous polynomial in (a,b) of degree $|E| - |S|$.

Proposition 9. *Let G be the net of Example 1 and $a < b$ integers. Then $\eta_v^{a,b}(G) > 0$ if and only if $b = -a$. Moreover, for $b = -a$ there are polynomials h_v such that*

$$q_D = \sum_{v \in V} h_v\, (x_v^2 - b^2).$$

Proof. Take the orientation $A = \{(t_1,t_2),(t_2,t_3),(t_1,t_3),(t_1,p_1),(t_2,p_2),(t_3,p_3)\}$. Then q_D has 318 terms, and a computer algebra computation gives

$$R_D = 4(a+b)^2\,\Phi_4 + 4(a-b)^2(a+b)\,\Phi_3 + 4(a+b)^2(a^2-3ab+b^2)\,\Phi_2,$$

where

$$\begin{aligned}
\Phi_4 &= t_1t_2p_1p_3 - t_1t_2p_2p_3 - t_1t_3p_1p_2 + t_1t_3p_2p_3 + t_2t_3p_1p_2 - t_2t_3p_1p_3,\\
\Phi_3 &= -t_1t_2p_1 + t_1t_2p_2 + t_1t_3p_1 - t_1t_3p_3 - t_2t_3p_2 + t_2t_3p_3\\
&\quad + t_1p_1p_2 - t_1p_1p_3 - t_2p_1p_2 + t_2p_2p_3 + t_3p_1p_3 - t_3p_2p_3,\\
\Phi_2 &= -t_1p_2 + t_1p_3 + t_2p_1 - t_2p_3 - t_3p_1 + t_3p_2
\end{aligned}$$

(here each vertex name stands for its variable). These are 24 distinct multilinear monomials, and the gcd of the coefficients of R_D is $4(a+b)$. If $a+b \neq 0$, the coefficient $4(a+b)^2$ of $t_1t_2p_1p_3$ is nonzero, so $R_D(a,b;x) \neq 0$ and, by (16), $\eta_v^{a,b}(G) = 0$. If $a+b=0$, then $R_D(a,b;x) = 0$, so $\eta_v^{a,b}(G) > 0$; moreover $(x_v-a)(x_v-b) = x_v^2 - b^2$, and Lemma 5 gives the polynomials h_v. (The computation yields cofactors with 112, 100, 76, 48, 39 and 24 terms for t_1,t_2,t_3,p_1,p_2,p_3.) □

Remark 4. For $b=1$, the ideal $\langle x_v^2 - 1 : v \in V\rangle$ is the ideal I_{n2} of the square roots of unity $\{-1,1\}$. Thus Proposition 9 is a negative certificate showing that the net is not "additive colorable with the square roots of unity", although it is additive 2-colorable. Furthermore, $(-1,1)$ is, up to scaling, the only pair of labels for which this happens.

Remark 5. In general, let $g_G(a,b)$ be the gcd of the coefficients of R_D (it does not depend on D up to sign, since reversing an arc changes the sign of q_D). If $R_D = 0$, then $\eta_v^{a,b}(G) > 0$ for every $a < b$ (e.g. $G = K_3$). Otherwise, as the coefficients are homogeneous binary forms, they factor over $\mathbb{C}$ into linear forms, and their common zeros are exactly the zeros of g_G. Therefore the pairs $a < b$ with $\eta_v^{a,b}(G) > 0$ are those with $g_G(a,b) = 0$, and they form finitely many ratios $a:b$.

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